1. Create an Address class with data members - HouseNo, Colony, Area, City and Pincode. Implement followings:

- Default constructor
- Parameterized constructor
- Accept Function for data input
- Display function for data display
- Separate header and implementation files
- Write function to find whether two addresses are same or different.
- Overlaod == operator

Address.h

```
#ifndef ADDRESS_H
#define ADDRESS_H

#include <iostream>
#include <string>
using namespace std;

class Address
{
private:
   int houseNo;
   string colony;
   string area;
   string city;
   int pincode;

public:
   // constructors
   Address();
   Address(int h, string c, string a, string ci, int p);

   // member functions
   void accept();
   void display();
   bool isSame(Address a);

   // operator overloading
   bool operator==(Address a);
};

#endif
```

Address.cpp

```
#include "Address.h"

// Default constructor
Address::Address()
{
   houseNo = 0;
   colony = "";
   area = "";
   city = "";
   pincode = 0;
}

// Parameterized constructor
Address::Address(int h, string c, string a, string ci, int p)
{
   houseNo = h;
   colony = c;
   area = a;
   city = ci;
   pincode = p;
}

// Accept function
void Address::accept()
{
   cout << "Enter House No: ";
   cin >> houseNo;

   cout << "Enter Colony: ";
   cin >> colony;

   cout << "Enter Area: ";
   cin >> area;

   cout << "Enter City: ";
   cin >> city;

   cout << "Enter Pincode: ";
   cin >> pincode;
}
```

```
// Display function
void Address::display()
{
   cout << "House No: " << houseNo << endl;
   cout << "Colony: " << colony << endl;
   cout << "Area: " << area << endl;
   cout << "City: " << city << endl;
   cout << "Pincode: " << pincode << endl;
}

// Compare addresses
bool Address::isSame(Address a)
{
   if (houseNo == a.houseNo &&
       colony == a.colony &&
       area == a.area &&
       city == a.city &&
       pincode == a.pincode)
       return true;

   return false;
}

// Operator overloading ==
bool Address::operator==(Address a)
{
   return isSame(a);
}
```

main.cpp

```
#include "Address.h"
#include <iostream>
using namespace std;

int main()
{
   Address a1, a2;

   cout << "Enter Address 1" << endl;
   a1.accept();

   cout << "\nEnter Address 2" << endl;
```

```cpp
    a2.accept();

    cout << "\nAddress 1" << endl;
    a1.display();

    cout << "\nAddress 2" << endl;
    a2.display();

    if (a1.isSame(a2))
        cout << "\nAddresses are SAME" << endl;
    else
        cout << "\nAddresses are DIFFERENT" << endl;

    if (a1 == a2)
        cout << "(Using operator==) Addresses are SAME" << endl;
    else
        cout << "(Using operator==) Addresses are DIFFERENT" << endl;

    return 0;
}
```

```
PROBLEMS    OUTPUT    TERMINAL    PORTS    DEBUG CONSOLE

~/Documents/CDAC/C++/doc/SLA1
g++ main.cpp Address.cpp -o address

~/Documents/CDAC/C++/doc/SLA1
./address
Enter Address 1
Enter House No: 983
Enter Colony: Vayu Nagar
Enter Area: Enter City: Pune
Enter Pincode: 411014

Enter Address 2
Enter House No: 338
Enter Colony: Gandhi Nagar
Enter Area: Enter City: Delhi
Enter Pincode: 110010

Address 1
House No: 983
Colony: Vayu
Area: Nagar
City: Pune
Pincode: 411014

Address 2
House No: 338
Colony: Gandhi
Area: Nagar
City: Delhi
Pincode: 110010

Addresses are DIFFERENT
(Using operator==) Addresses are DIFFERENT
```

2. Create Payslip class with data members
EmpNo
EmpName
DA
HRA
BasicSalary
Implement followings
• Default constructor
• Parameterized constructor
• Accept Function for data input
• Display function for data display
• Function to calculate salary from DA, HRA and Basic Salary
• Separate header and implementation files

Payslip.h

```
#ifndef PAYSLIP_H
#define PAYSLIP_H

#include <iostream>
#include <string>
using namespace std;

class Payslip
{
private:
   int empNo;
   string empName;
   float DA;
   float HRA;
   float basicSalary;

public:
   // constructors
   Payslip();
   Payslip(int no, string name, float da, float hra, float basic);

   // member functions
   void accept();
   void display();
   float calculateSalary();
};

#endif
```

Payslip.cpp

```
#include "Payslip.h"

// Default constructor
Payslip::Payslip()
{
   empNo = 0;
   empName = "";
   DA = 0;
   HRA = 0;
   basicSalary = 0;
}

// Parameterized constructor
Payslip::Payslip(int no, string name, float da, float hra, float basic)
{
   empNo = no;
   empName = name;
   DA = da;
   HRA = hra;
   basicSalary = basic;
}

// Accept function
void Payslip::accept()
{
   cout << "Enter Employee Number: ";
   cin >> empNo;

   cin.ignore();

   cout << "Enter Employee Name: ";
   getline(cin, empName);

   cout << "Enter Basic Salary: ";
   cin >> basicSalary;

   cout << "Enter DA: ";
   cin >> DA;

   cout << "Enter HRA: ";
   cin >> HRA;
```

```
}

// Salary calculation
float Payslip::calculateSalary()
{
   return basicSalary + DA + HRA;
}

// Display function
void Payslip::display()
{
   cout << "\nEmployee Number: " << empNo << endl;
   cout << "Employee Name: " << empName << endl;
   cout << "Basic Salary: " << basicSalary << endl;
   cout << "DA: " << DA << endl;
   cout << "HRA: " << HRA << endl;
   cout << "Total Salary: " << calculateSalary() << endl;
}
```

2main.cpp

```
#include "Payslip.h"
#include <iostream>
using namespace std;

int main()
{
   Payslip p1;

   cout << "Enter Employee Details\n";
   p1.accept();

   cout << "\nEmployee Payslip\n";
   p1.display();

   return 0;
}
```

O/P

```
PROBLEMS    OUTPUT    TERMINAL    PORTS    DEBUG CONSOLE

~/Documents/CDAC/C++/doc/SLA1
g++ 2main.cpp Payslip.cpp -o payslip

~/Documents/CDAC/C++/doc/SLA1
./payslip
Enter Employee Details
Enter Employee Number: 8987678
Enter Employee Name: Sudhanshu Singh
Enter Basic Salary: 98453
Enter DA: 23098
Enter HRA: 43500

Employee Payslip

Employee Number: 8987678
Employee Name: Sudhanshu Singh
Basic Salary: 98453
DA: 23098
HRA: 43500
Total Salary: 165051
```

3. Create MobilePhone class with data members
ModelNo
Price
Manufacturer
Quantity
Implement followings
• Default constructor
• Parameterized constructor
• Accept Function for data input
• Display function for data display
• Function to search MobilePhone model availability
• Write menu driven program with following menus
o Add Mobile
o Display List of available mobiles
o Find out quantity of particular mobile type
o Find out availability of Particular model

MobilePhone.h

```
#ifndef MOBILEPHONE_H
#define MOBILEPHONE_H

#include <iostream>
#include <string>
using namespace std;

class MobilePhone
{
private:
   string modelNo;
   float price;
   string manufacturer;
   int quantity;

public:
   // constructors
   MobilePhone();
   MobilePhone(string m, float p, string man, int q);

   // member functions
   void accept();
   void display();

   string getModelNo();
   int getQuantity();
};

#endif
```

MobilePhone.cpp

```
#include "MobilePhone.h"

// Default constructor
MobilePhone::MobilePhone()
{
   modelNo = "";
   price = 0;
   manufacturer = "";
   quantity = 0;
```

```
}

// Parameterized constructor
MobilePhone::MobilePhone(string m, float p, string man, int q)
{
   modelNo = m;
   price = p;
   manufacturer = man;
   quantity = q;
}

// Accept mobile details
void MobilePhone::accept()
{
   cout<<"Enter Model No: ";
   cin>>modelNo;

   cout<<"Enter Price: ";
   cin>>price;

   cout<<"Enter Manufacturer: ";
   cin>>manufacturer;

   cout<<"Enter Quantity: ";
   cin>>quantity;
}

// Display mobile details
void MobilePhone::display()
{
   cout<<"Model No: "<<modelNo<<endl;
   cout<<"Price: "<<price<<endl;
   cout<<"Manufacturer: "<<manufacturer<<endl;
   cout<<"Quantity: "<<quantity<<endl;
   cout<<"--------------------------"<<endl;
}

// Getter functions
string MobilePhone::getModelNo()
{
   return modelNo;
}
```

```
int MobilePhone::getQuantity()
{
   return quantity;
}
```

3main.cpp

```
#include "MobilePhone.h"
#include <iostream>
using namespace std;

int main()
{
   MobilePhone mobiles[50];
   int count = 0;
   int choice;

   do
   {
       cout<<"\n---- Mobile Management System ----\n";
       cout<<"1. Add Mobile\n";
       cout<<"2. Display List of Mobiles\n";
       cout<<"3. Find Quantity of Particular Mobile\n";
       cout<<"4. Check Availability of Model\n";
       cout<<"5. Exit\n";
       cout<<"Enter choice: ";
       cin>>choice;

       switch(choice)
       {
           case 1:
               mobiles[count].accept();
               count++;
               break;

           case 2:
               for(int i=0;i<count;i++)
                   mobiles[i].display();
               break;

           case 3:
```

```
        {
            string model;
            cout<<"Enter model number: ";
            cin>>model;

            for(int i=0;i<count;i++)
            {
                if(mobiles[i].getModelNo()==model)
                    cout<<"Quantity: "<<mobiles[i].getQuantity()<<endl;
            }
            break;
        }

        case 4:
        {
            string model;
            bool found=false;

            cout<<"Enter model number: ";
            cin>>model;

            for(int i=0;i<count;i++)
            {
                if(mobiles[i].getModelNo()==model)
                {
                    cout<<"Model Available\n";
                    found=true;
                }
            }

            if(!found)
                cout<<"Model Not Available\n";

            break;
        }

        case 5:
            cout<<"Exiting...\n";
            break;

        default:
            cout<<"Invalid choice\n";
```

```
        }

    } while(choice!=5);

    return 0;
}
```

O/P

```
~/Documents/CDAC/C++/doc/SLA1
clear
~/Documents/CDAC/C++/doc/SLA1
g++ 3main.cpp MobilePhone.cpp -o mobile

~/Documents/CDAC/C++/doc/SLA1
./mobile

---- Mobile Management System ----
1. Add Mobile
2. Display List of Mobiles
3. Find Quantity of Particular Mobile
4. Check Availability of Model
5. Exit
Enter choice: 1
Enter Model No: Iphone 15
Enter Price: Enter Manufacturer: Apple
Enter Quantity: 5

---- Mobile Management System ----
1. Add Mobile
2. Display List of Mobiles
3. Find Quantity of Particular Mobile
4. Check Availability of Model
5. Exit
Enter choice: 2
Model No: Iphone
Price: 15
Manufacturer: Apple
Quantity: 5
```